PYTHON BASICS

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Example 1: Hello, World!

```
print("Hello, Machine Learning!")
```

Output

```
Hello, Machine Learning!
```

Concept: Displaying output using print().

Example 2: Variables

```
name = "Swati"
age = 35
marks = 92.5

print("Name:", name)
print("Age:", age)
print("Marks:", marks)
```

Output

Name: Swati
Age: 35
Marks: 92.5

Concept: Variables can store text, integers, and decimal values.

Example 3: User Input

```
name = input("Enter your name: ")

print("Welcome", name)
```

Sample Output

Enter your name: Unnati
Welcome Unnati

Concept: Reading input from the keyboard.

Example 4: Simple Calculator

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

print("Addition =", a + b)
print("Subtraction =", a - b)
print("Multiplication =", a * b)
print("Division =", a / b)
```

Sample Output

Enter first number: 20
Enter second number: 5

Addition = 25
Subtraction = 15
Multiplication = 100
Division = 4.0

Concept: Arithmetic operators.

Example 5: If-Else Statement

```
marks = int(input("Enter marks: "))

if marks >= 40:
    print("Pass")
```

```
else:
    print("Fail")
```

Output

```
Enter marks: 75

Pass
```

Concept: Decision making.

Example 6: Find Even or Odd

```
number = int(input("Enter a number: "))

if number % 2 == 0:
    print("Even Number")
else:
    print("Odd Number")
```

Output

```
Enter a number: 18

Even Number
```

Concept: Modulus operator and conditions.

Example 7: For Loop

```
for i in range(1, 6):
    print(i)
```

Output

```
1
2
3
4
5
```

Concept: Repeating tasks.

Example 8: Multiplication Table

```
number = int(input("Enter a number: "))

for i in range(1, 11):
    print(number, "x", i, "=", number * i)
```

Output

5 x 1 = 5
5 x 2 = 10
...
5 x 10 = 50

Concept: Loops.

Example 9: While Loop

```
count = 1

while count <= 5:
    print("Machine Learning")
    count += 1
```

Output

Machine Learning
Machine Learning
Machine Learning
Machine Learning
Machine Learning

Concept: Repetition using a condition.

Example 10: List

```
languages = ["Python", "Java", "C++", "R"]

print(languages)

print("First Language:", languages[0])

print("Total Languages:", len(languages))
```

Output

['Python', 'Java', 'C++', 'R']

First Language: Python

Total Languages: 4

Concept: Storing multiple values.

Example 11: Loop Through a List

```
fruits = ["Apple", "Banana", "Orange"]

for fruit in fruits:
    print(fruit)
```

Output

```
Apple
Banana
Orange
```

Concept: Iterating over a list.

Example 12: Dictionary

```
student = {
    "Name": "Rahul",
    "Age": 20,
    "Course": "Cyber Security"
}

print(student)

print("Student Name:", student["Name"])
```

Output

```
{'Name': 'Rahul', 'Age': 20, 'Course': 'Cyber Security'}

Student Name: Rahul
```

Concept: Key-value pairs.

Example 13: Function

```
def greet(name):
    print("Welcome", name)

greet("Unnati")
greet("Swati")
```

Output

```
Welcome Unnati
Welcome Swati
```

Concept: Creating reusable code.

Example 14: Find Maximum Number

```
numbers = [15, 28, 7, 42, 19]

print("Maximum Number:", max(numbers))
```

Output

```
Maximum Number: 42
```

Concept: Built-in functions.

Example 15: Basic NumPy Example

```
import numpy as np

numbers = np.array([10, 20, 30, 40])

print(numbers)

print("Mean =", np.mean(numbers))
```

Output

```
[10 20 30 40]

Mean = 25.0
```

Concept: Numerical computing for Machine Learning.

Example 16: Basic Pandas Example

```
import pandas as pd

student = {
   "Name": ["Amit", "Neha", "Riya"],
   "Marks": [85, 90, 78]
}

df = pd.DataFrame(student)

print(df)
```

Output

```
  Name  Marks
0  Amit    85
1  Neha    90
2  Riya    78
```

Concept: Creating and viewing tabular data.

Example 17: Basic Matplotlib Example

```
import matplotlib.pyplot as plt

marks = [70, 85, 90, 75, 95]

plt.plot(marks)
plt.title("Student Marks")
plt.xlabel("Student")
plt.ylabel("Marks")
plt.show()
```

Output: A simple line graph showing students' marks.

Concept: Data visualization.

Example 18: First Machine Learning Program

```
from sklearn.tree import DecisionTreeClassifier

# Features: Hours Studied
X = [[1], [2], [3], [4], [5]]

# Labels: Pass (1) or Fail (0)
y = [0, 0, 1, 1, 1]

model = DecisionTreeClassifier()
model.fit(X, y)

prediction = model.predict([[2.5]])

print("Prediction:", prediction)
```

Output

```
Prediction: [1]
```

Concept: A simple supervised learning example where the model learns from labeled data and predicts whether a student is likely to pass based on study hours.
